

What is Kafka and its Installation and Guide in Ubuntu

Kafka is an event-streaming platform or cluster server. Producers write or push records to topics, brokers store the records, and consumers read or pull them. It uses push-pull techniques. Kafka can handle multiple sources and multiple destinations.

Installation:

1. Open a terminal, Check java version: `java -version`
2. Download Kafka from: <https://kafka.apache.org/community/downloads/>
3. Make a dedicated dir for Kafka, move the downloaded kafka zip file to that folder.
4. Unzip the folder and enter there

```
$tar -xzf kafka_2.13-4.3.1.tgz
```

```
$ cd kafka_2.13-4.3.1
```

```
$ ls
```

`bin config libs LICENSE licenses logs NOTICE site-docs`

5. Initialize Kafka:

```
$ KAFKA_CLUSTER_ID="$(bin/kafka-storage.sh random-uuid)"
```

```
$ bin/kafka-storage.sh format \
```

```
--standalone \
```

```
-t $KAFKA_CLUSTER_ID \
```

```
-c config/server.properties
```

- 6 . Start Kafka environment:

```
$ bin/kafka-server-start.sh config/server.properties
```

Keep open this shell as long as you are working with kafka, so the server is connected. Let's say this is terminal 0

7. Open another new terminal, enter to that folder, for me it is: (Let's say this is terminal 1)

`~/Susmita/Projects/Data_Engg/Kafka/kafka_2.13-4.3.`

Next commands are:

```
bin/kafka-topics.sh
```

```
bin/kafka-console-producer.sh
```

```
bin/kafka-console-consumer.sh
```

```
bin/kafka-server-start.sh
```

8. Create topic: Create a topic

```
$ bin/kafka-topics.sh --bootstrap-server localhost:9092 --create --topic IronHackSR --partitions 1 --replication-factor 1
```

9. Inspect the topic

```
$ bin/kafka-topics.sh --bootstrap-server localhost:9092 --list
```

```
output: IronHackSR
```

```
$ bin/kafka-topics.sh --describe \
```

```
--topic IronHackSR \
```

```
--bootstrap-server localhost:9092
```

output :

Topic: IronHackSR

PartitionCount: 1

ReplicationFactor: 1

10. Make console producer and consumer

Open another terminal (Lets say this is terminal 2)

cd particular folder

```
$ bin/kafka-console-producer.sh \  
--bootstrap-server localhost:9092 \  
--topic IronHackSR
```

one should get >

means the producer is waiting for you to enter messages.

Enter ans messages:Hallo Kafka (then enter) This is my first message (Enter)
Test Kafka...

Terminal looks like:

```
> Hello Kafka  
> This is my first message  
> Test Kafka
```

11. Open another terminal (Lets say terminal 3), cd actual folder

```
bin/kafka-console-consumer.sh \  
--bootstrap-server localhost:9092 \  
--topic IronHackSR \  
--from-beginning
```

Terminal shows the messages entered in producer shell: - -from-beginning option tells the consumer to read existing messages from the beginning of the topic.

12. To test live communication:

enter any message in terminal 2, that will appear in terminal 3

Producer shell and the 2nd one is consumer shell

```
(base) susmita@roy:~/Susmita/Projects/Data_Engg/Kafka/kafka_2.13-4.3.1$ bin/kafka-console-producer.sh \  
--bootstrap-server localhost:9092 \  
--topic IronHackSR  
>  
>Hallo Kafka  
>My fisrt Kafka message  
>My topic name is IronHackSR  
>This is a new message  
>Testing Kafka consumer  
>Now I am testing to show somebody else  
>
```

```
(base) susmita@roy:~$ cd Susmita/Projects/Data_Engg/Kafka/kafka_2.13-4.3.1/  
(base) susmita@roy:~/Susmita/Projects/Data_Engg/Kafka/kafka_2.13-4.3.1$ bin/kafka-console-consumer.sh \  
--bootstrap-server localhost:9092 \  
--topic IronHackSR \  
--from-beginning  
The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.org/documentation/#consumer\_rebalance\_protocol  
  
Hallo Kafka  
My fisrt Kafka message  
My topic name is IronHackSR  
This is a new message  
Testing Kafka consumer  
Now I am testing to show somebody else
```